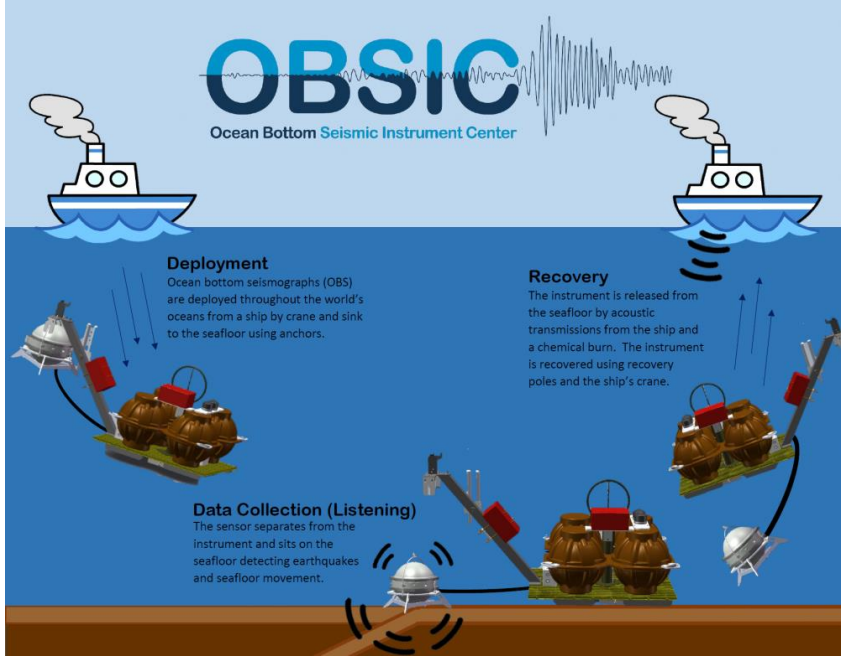
Appendix for ECS Poster

Gabrielle Arrieta

April 2026

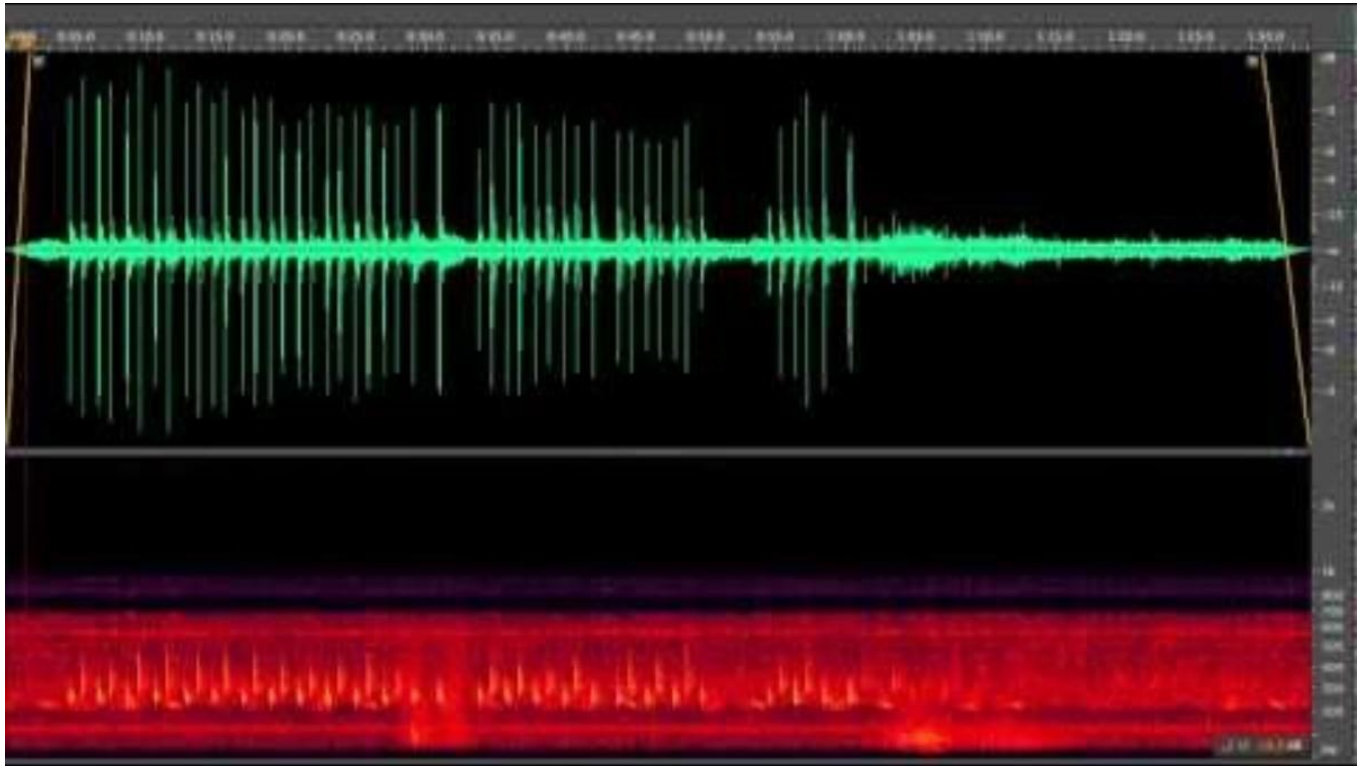
What is an Ocean Bottom Seismometer (OBS)?



- **Main objective:** understand the structure of the Earth's crust by measuring ground motion (e.g. earthquakes)
- **Common configuration:** 3-component OBS (4 sensors – 3 geophones + 1 hydrophone)
- **Type of array:** cabled or autonomous
- **Used by:** researchers, government agencies, commercial companies
- Passive or active source surveys (i.e. air guns)
- Some datasets are publicly available (i.e. through IRIS)

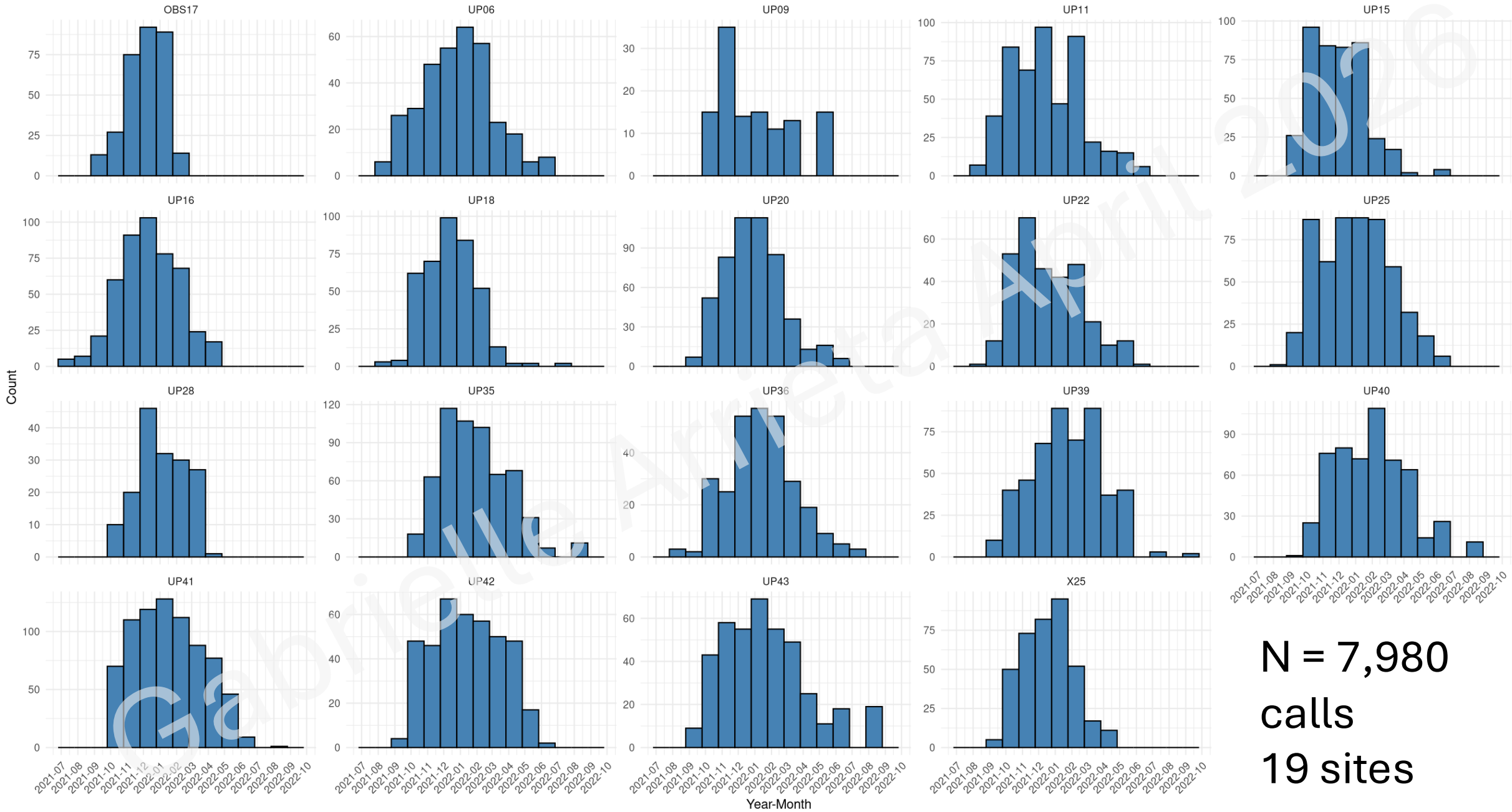
(WHOI, 2022)

Example of Fin Whale Calls in Seismic Data



CLICK HERE TO LISTEN !!

Manual Detections – Temporal Variability



N = 7,980 calls

19 sites

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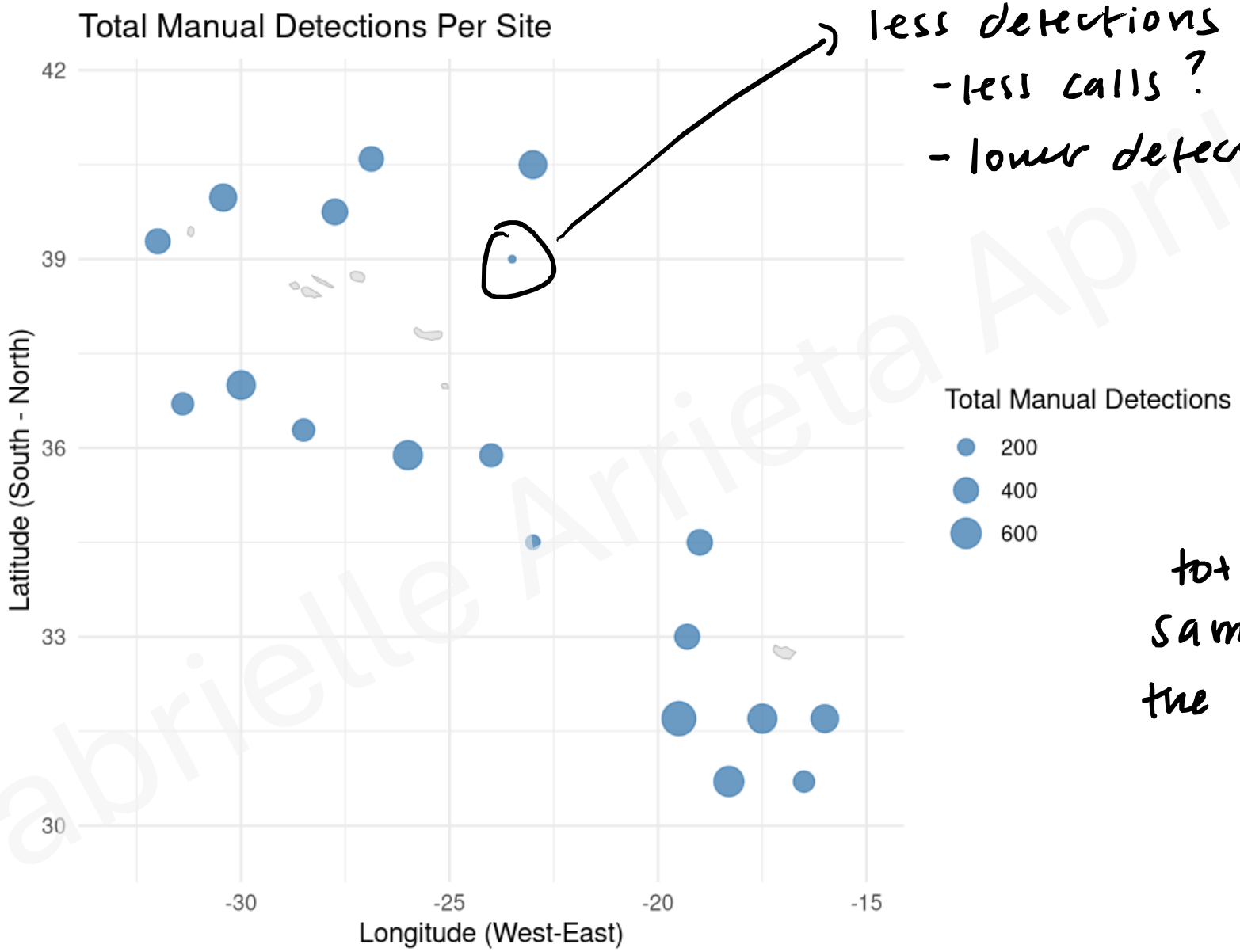
Manual Detections – Spatial Variability



less detections due to:
- less calls?
- lower detectability?

** each site was sampled the same amount
↓
total amount sampled ~3% of the entire dataset

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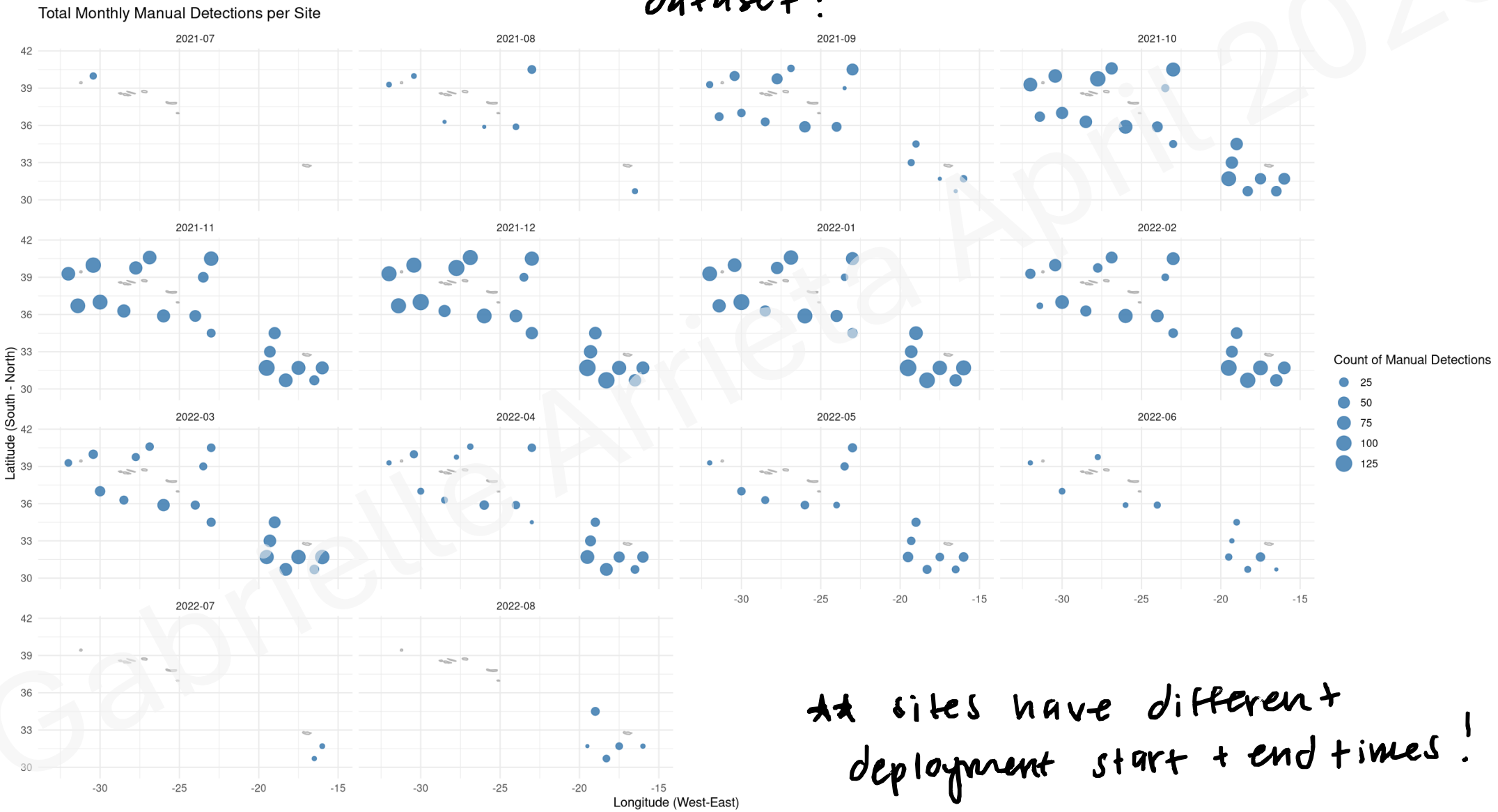
Manual Detections – Spatial and Temporal Variability

detections per each month of the dataset!



All sites have different deployment start + end times!

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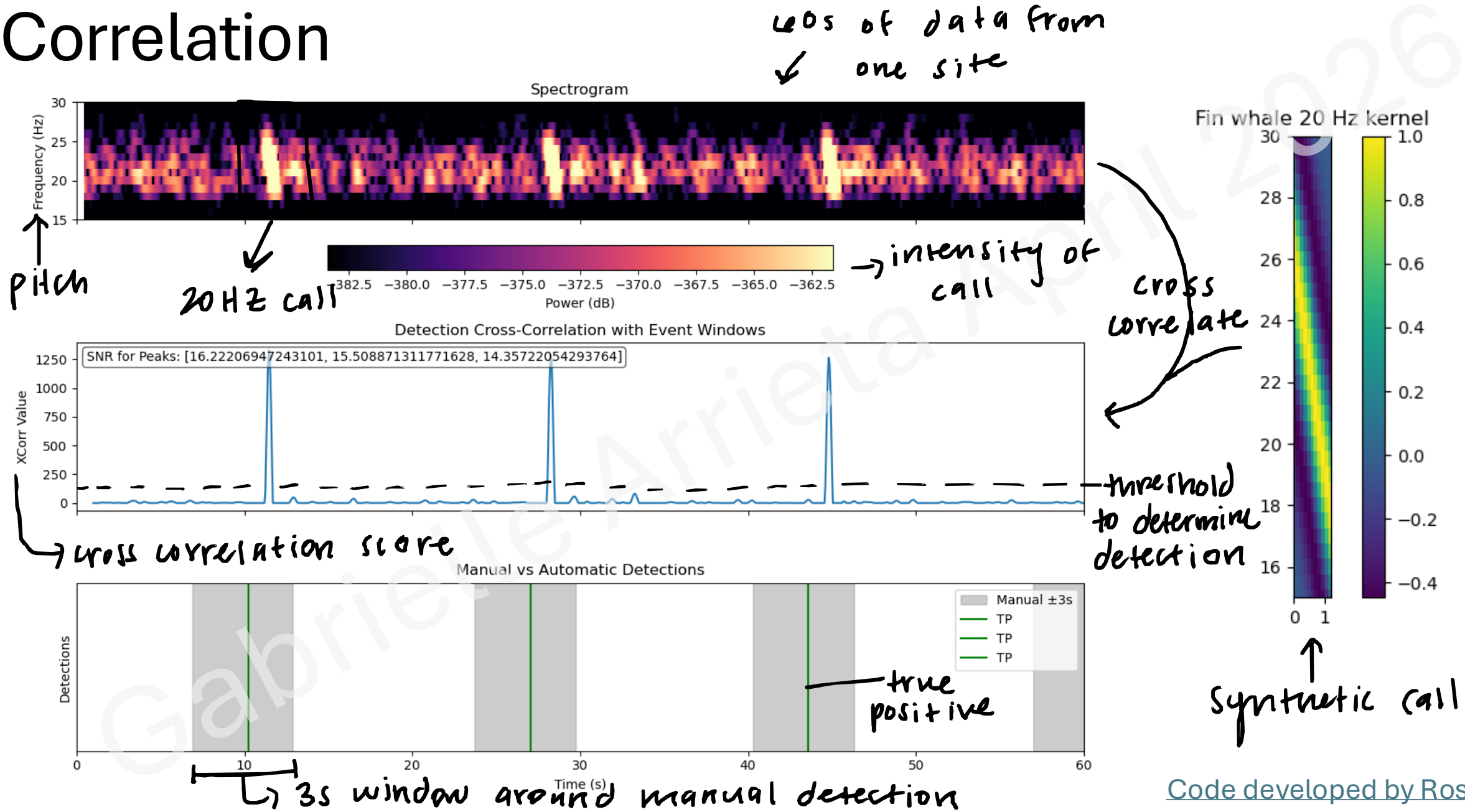
Automatic Detector – Spectrogram Cross Correlation



Code developed by Rose Hilmo

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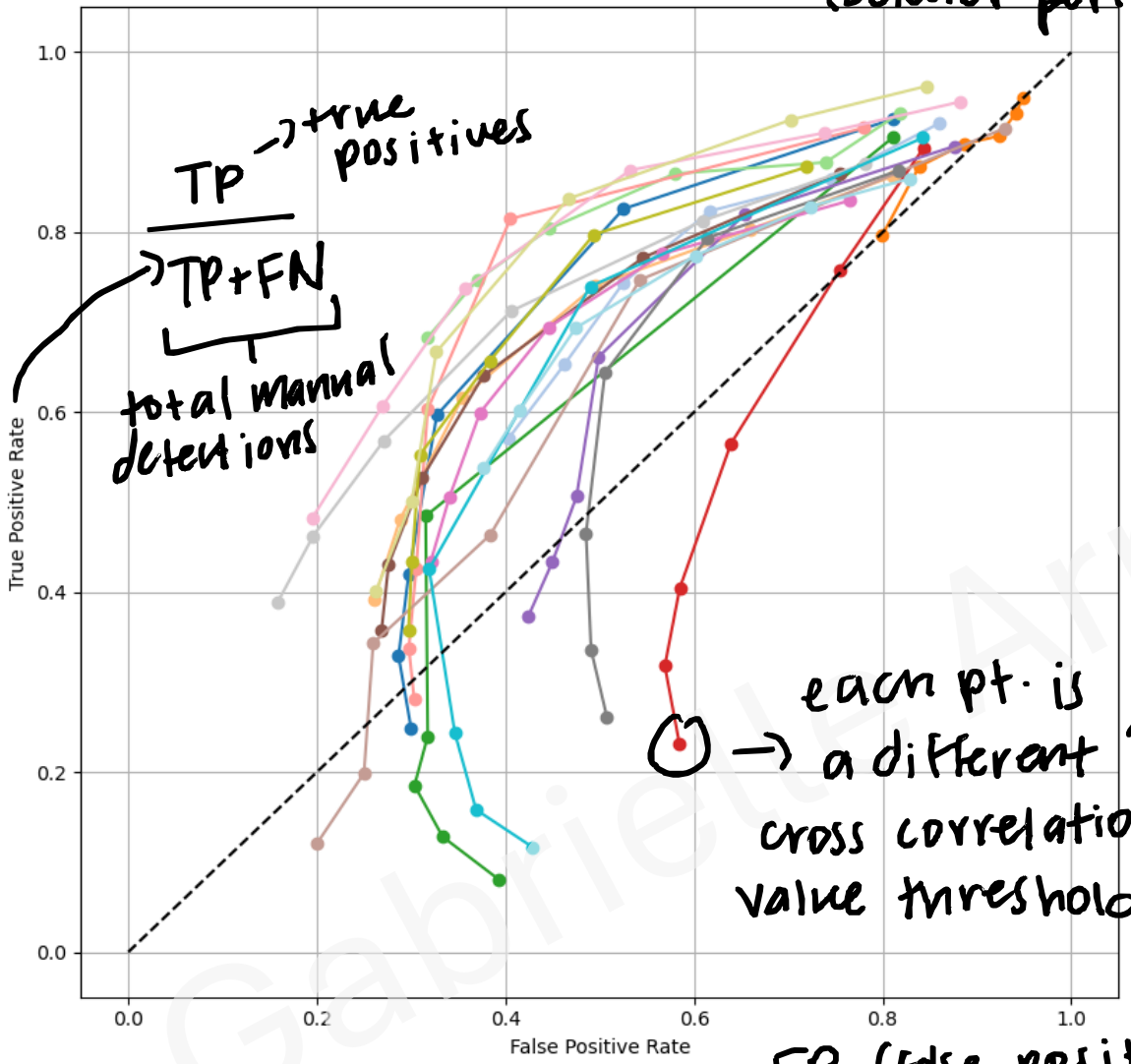
Sites Perform Differently !!!!

→ receiver operating curve (detector performance)

out of all the detections how many true ones did we get

how good detections are based on ground truth

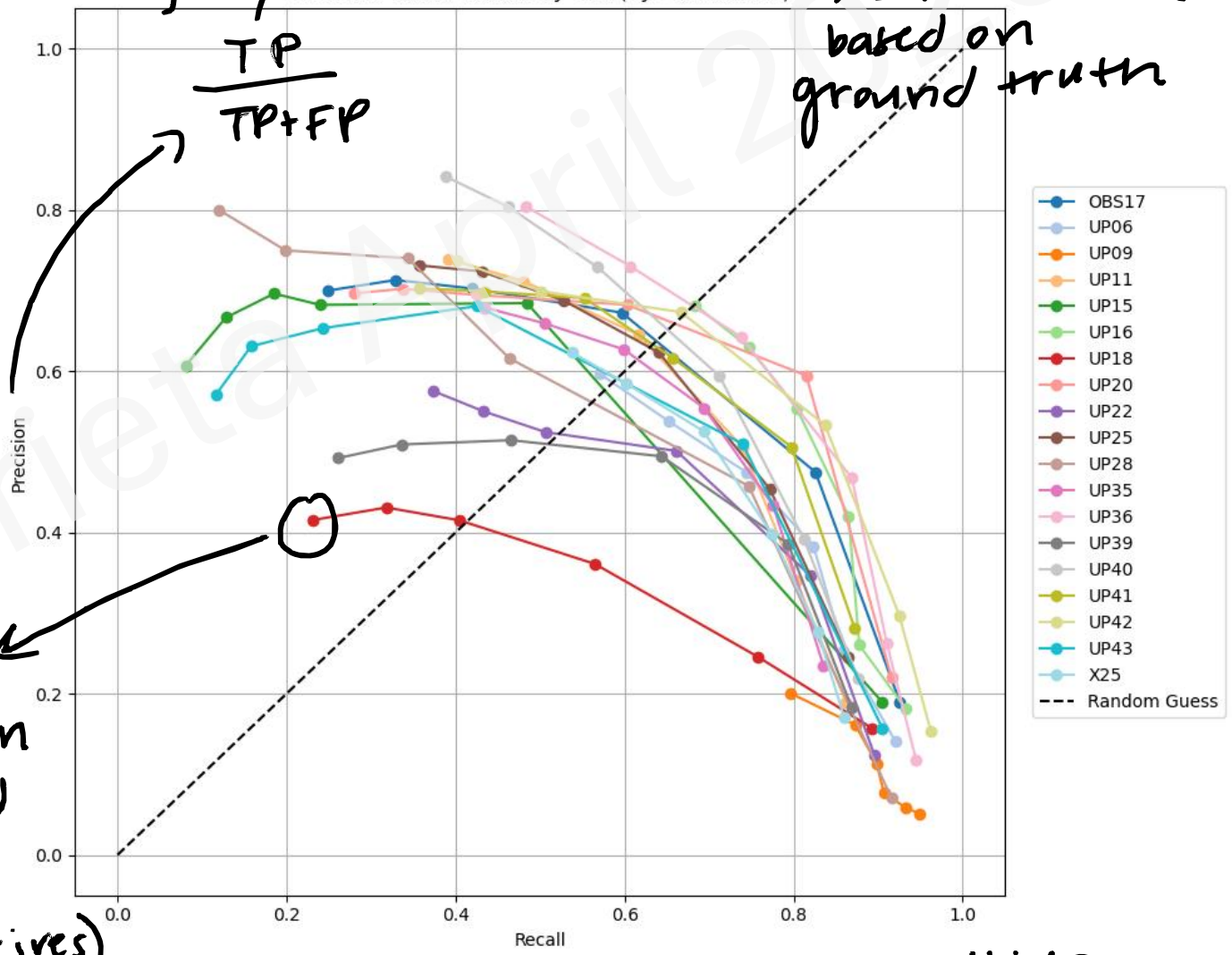


$\frac{TP}{TP+FN}$ → true positives

total manual detections

each pt. is a different cross correlation value threshold

↳ $\frac{FP}{FP+TP}$ (false positives) → total # of detections



$\frac{TP}{TP+FP}$

↳ also true positive rate

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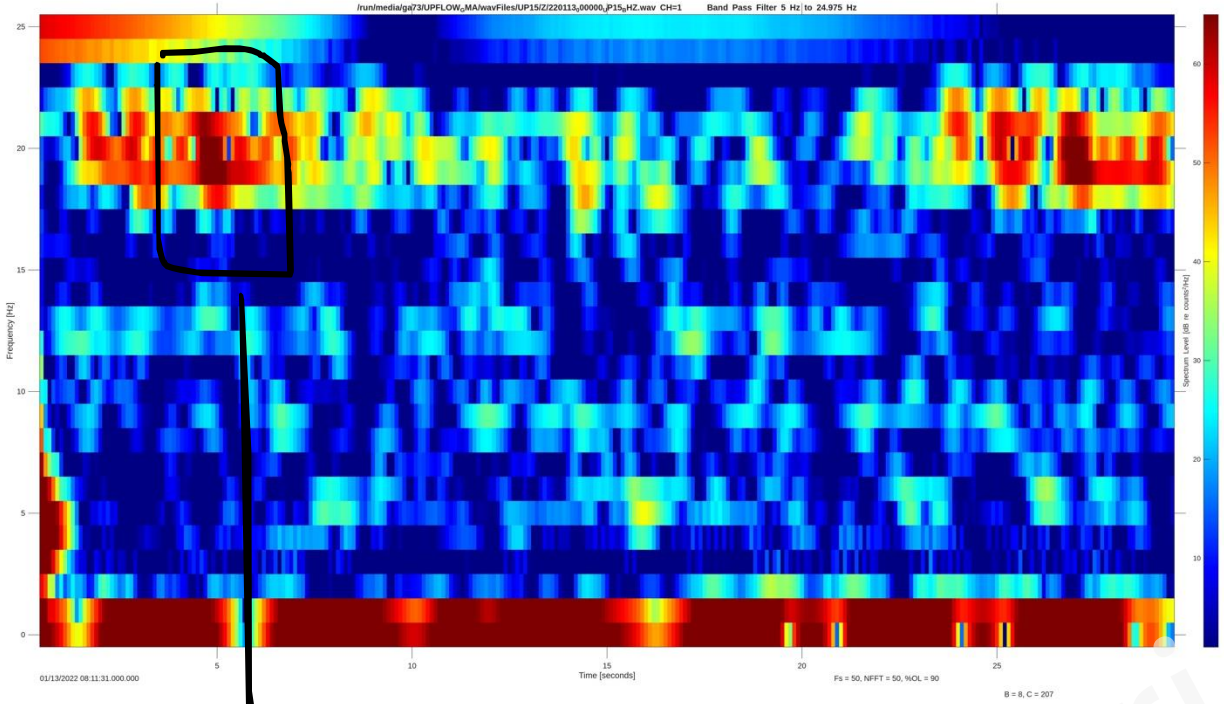
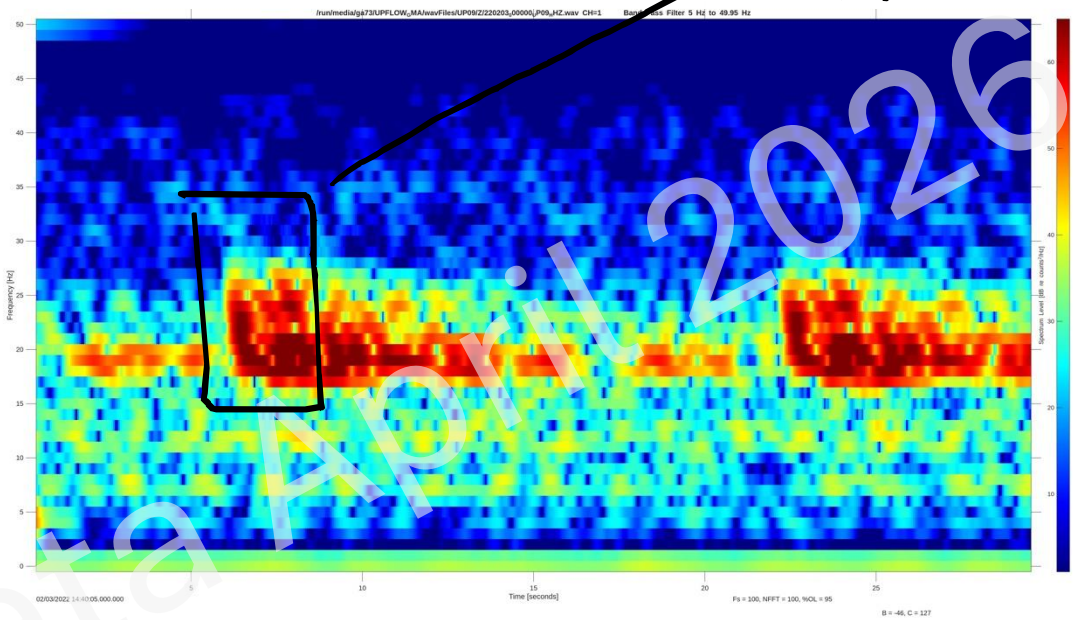
Fs = 50 (0-25 Hz)



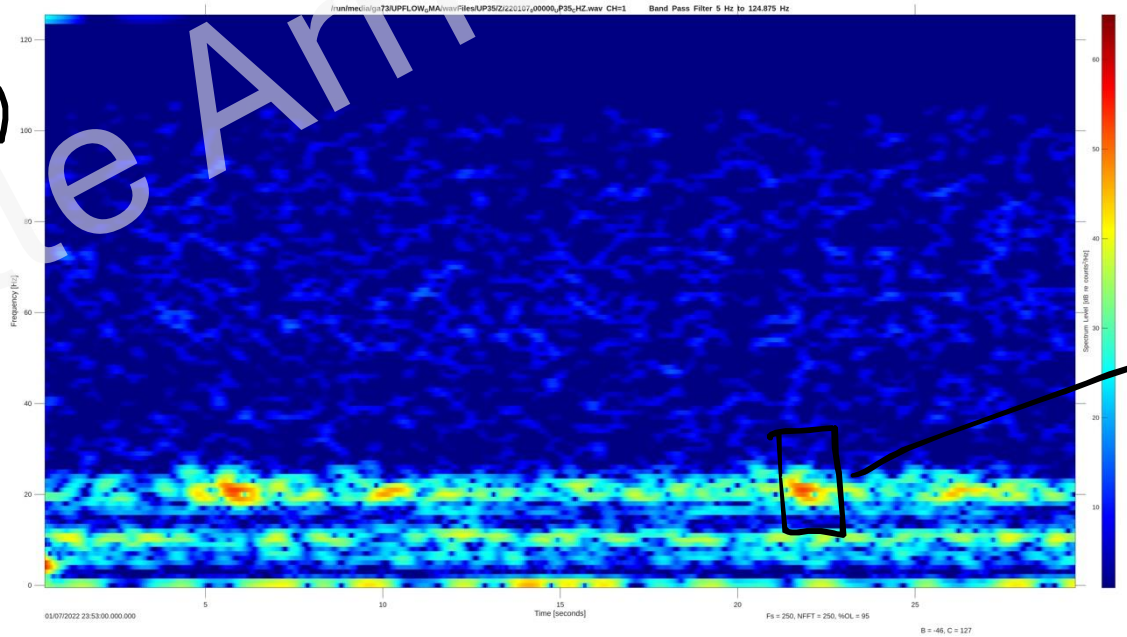
20 Hz call

Fs = 100 (0-50 Hz)

20 Hz call



Fs = 250 (0-125 Hz)



20 Hz call

More examples of how the calls look like with different sampling rates (Fs)

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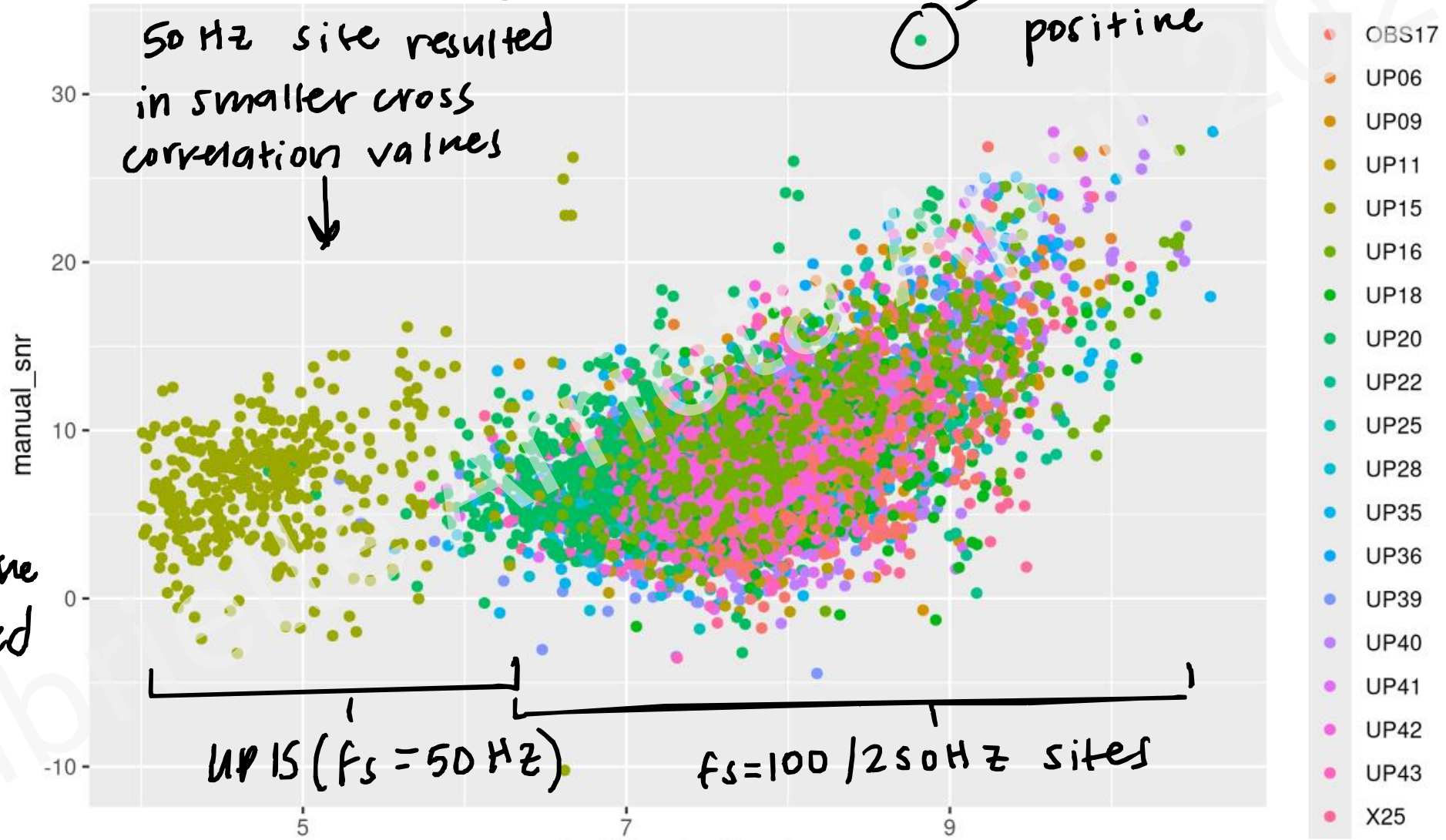
50 Hz Site vs the Other Sampling Rates – Cross Correlation Score

* same detector settings + synthetic call

each point is a true positive

50 Hz site resulted in smaller cross correlation values

signal to noise ratio → how intense the call is compared to background noise



→ cross correlation value

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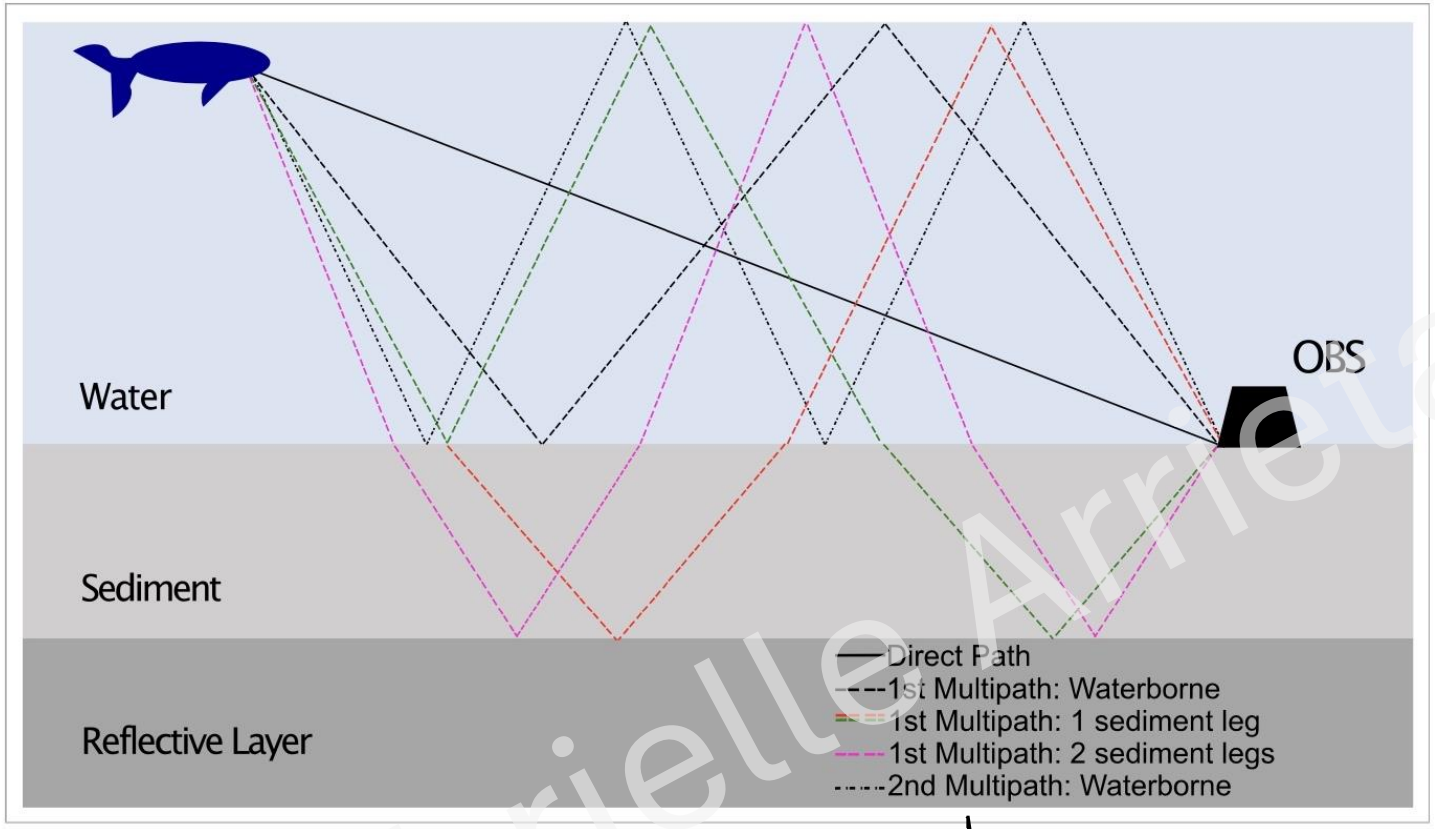
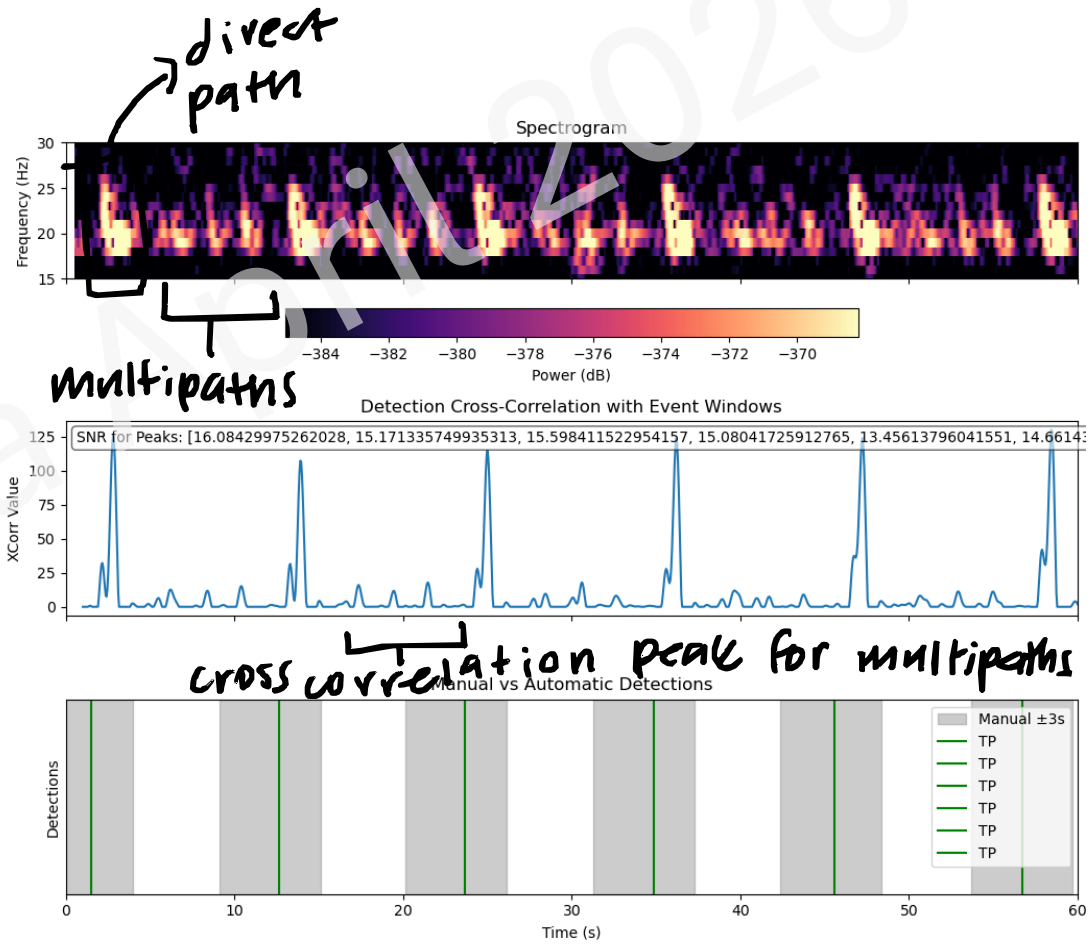
What are call echoes/multipaths?



# of paths dependent on seafloor bottom properties



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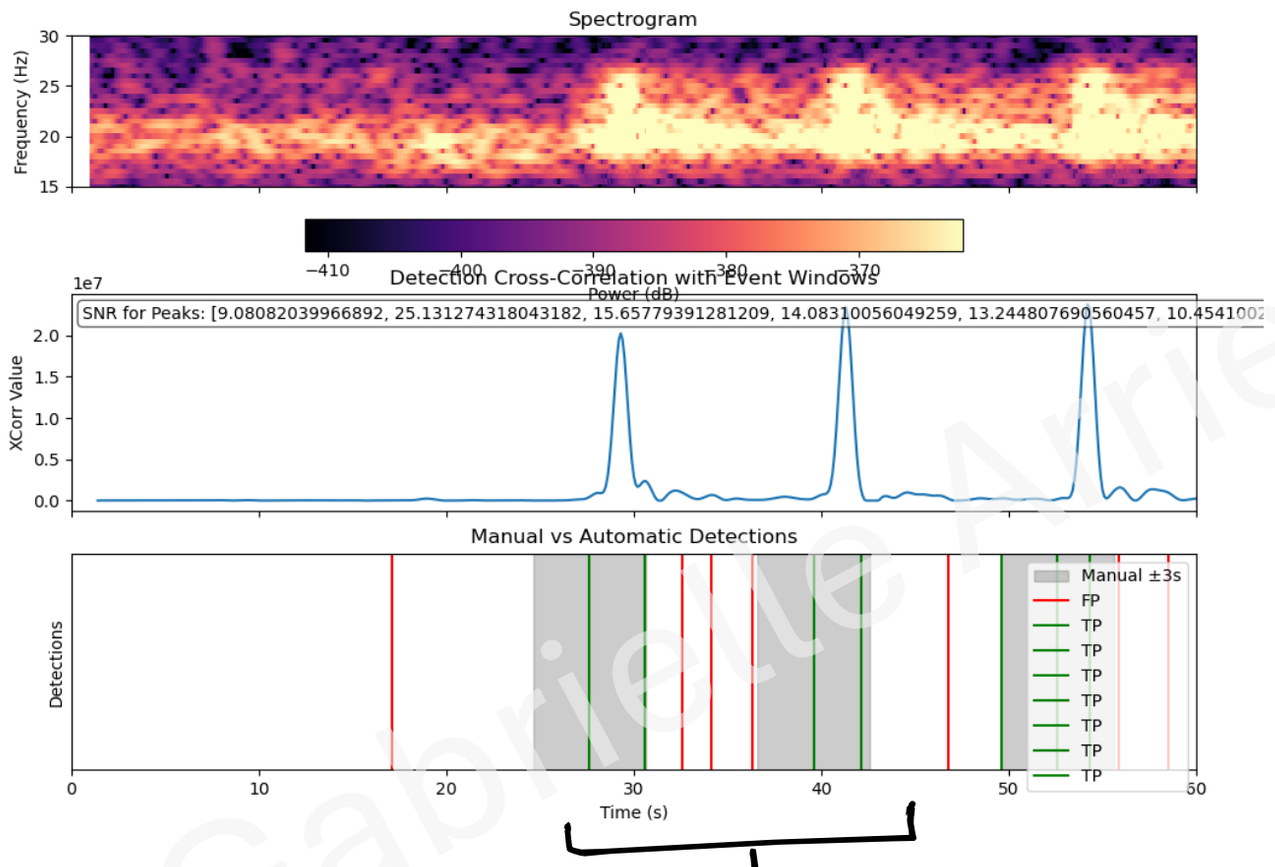
Other Complications – False Positives Caused by Multipaths

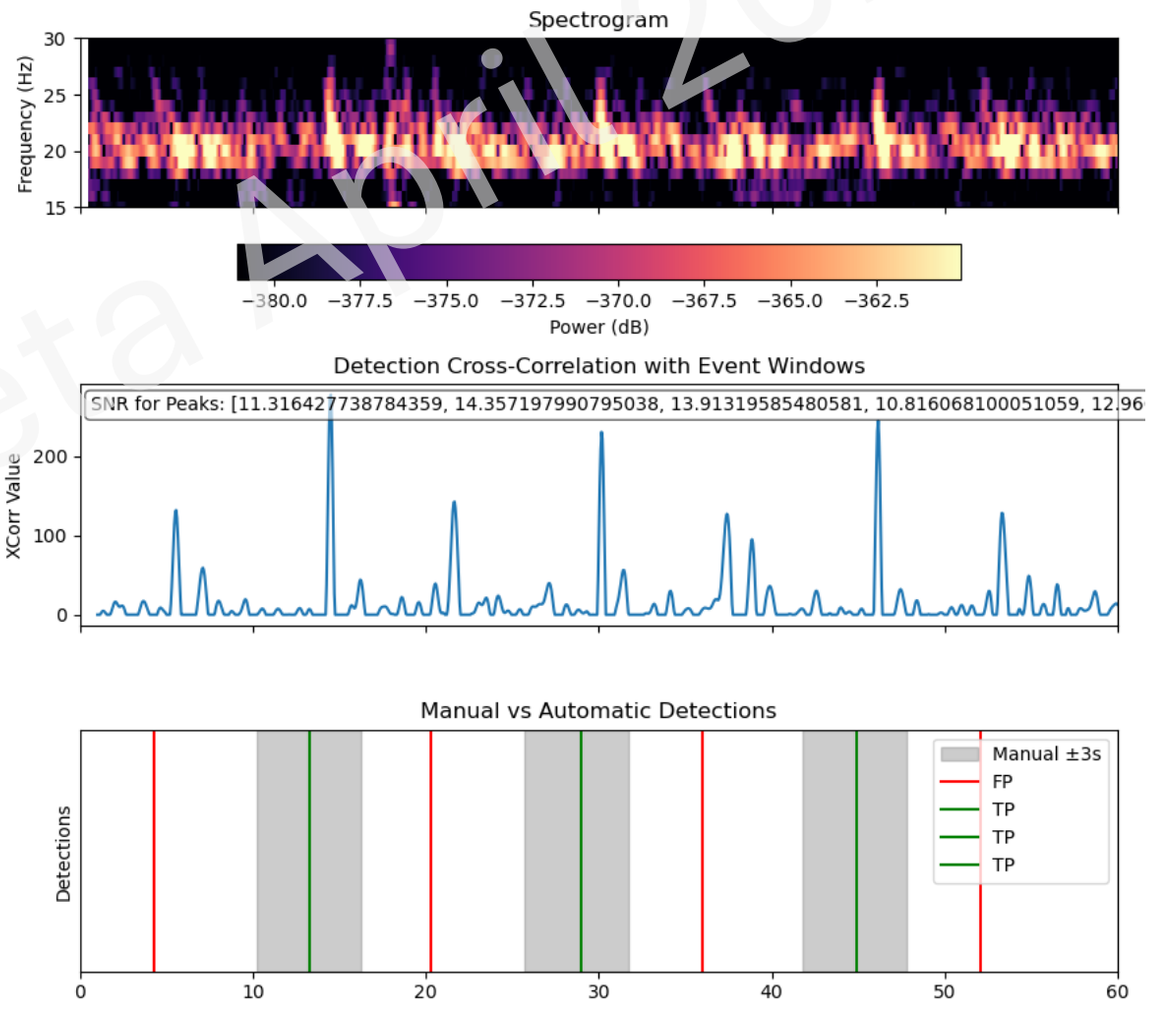
Buffer = 1s → number of seconds between automatic detections

→ Buffer = 7s

trade off between TP + FP



lots of false positives caused by multipaths



↳ multipaths still getting detected

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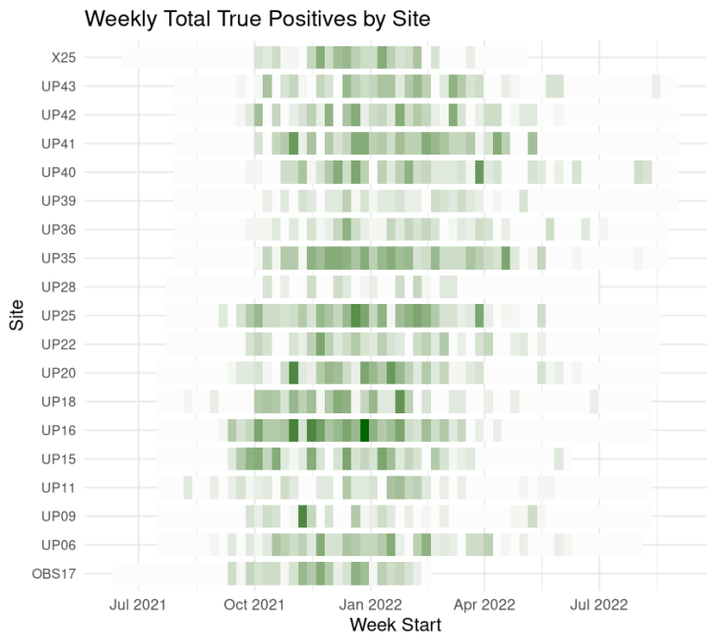
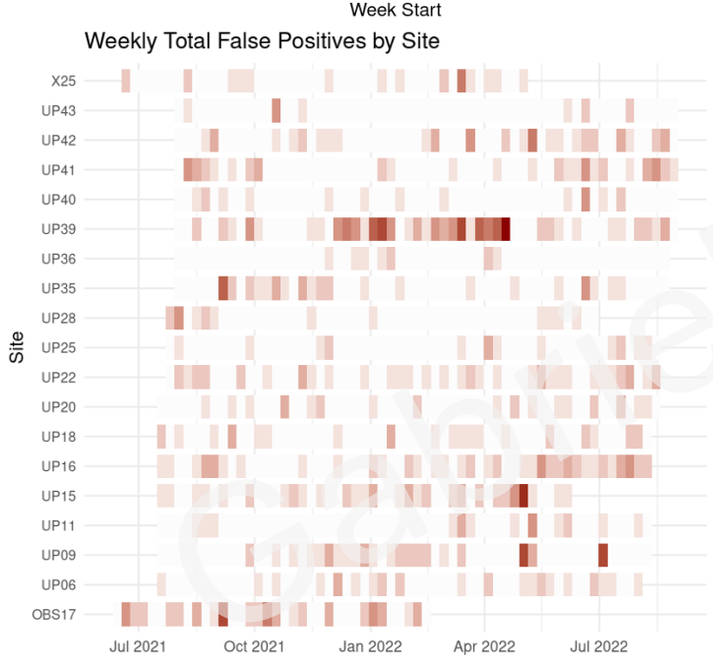
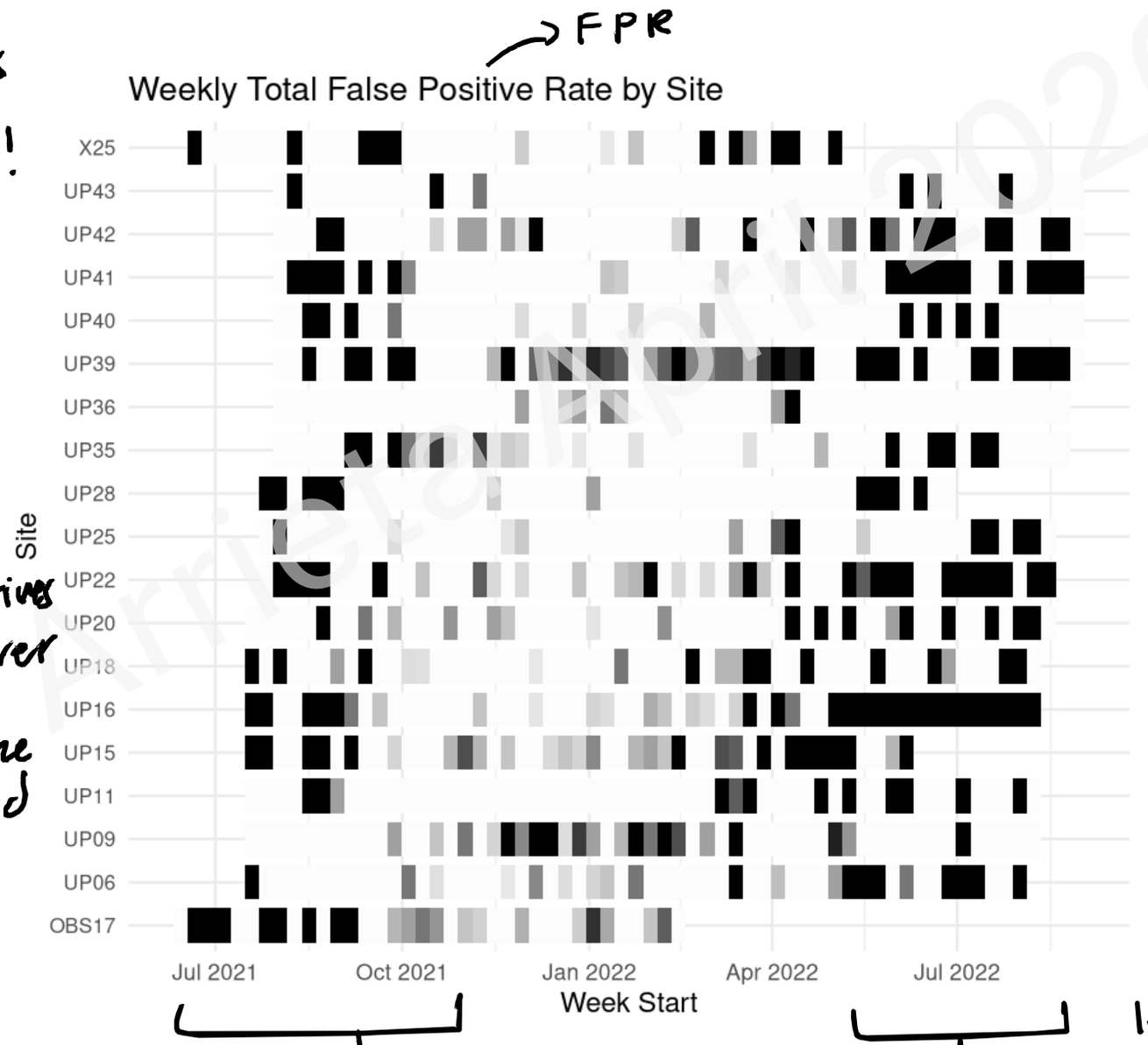
False Positive Rate Using Manual Detections



true positives are seasonal!



false positives spread over entire time period



$$\frac{FP}{TP+FP}$$

FPR higher at start and end of time period → less true positives!

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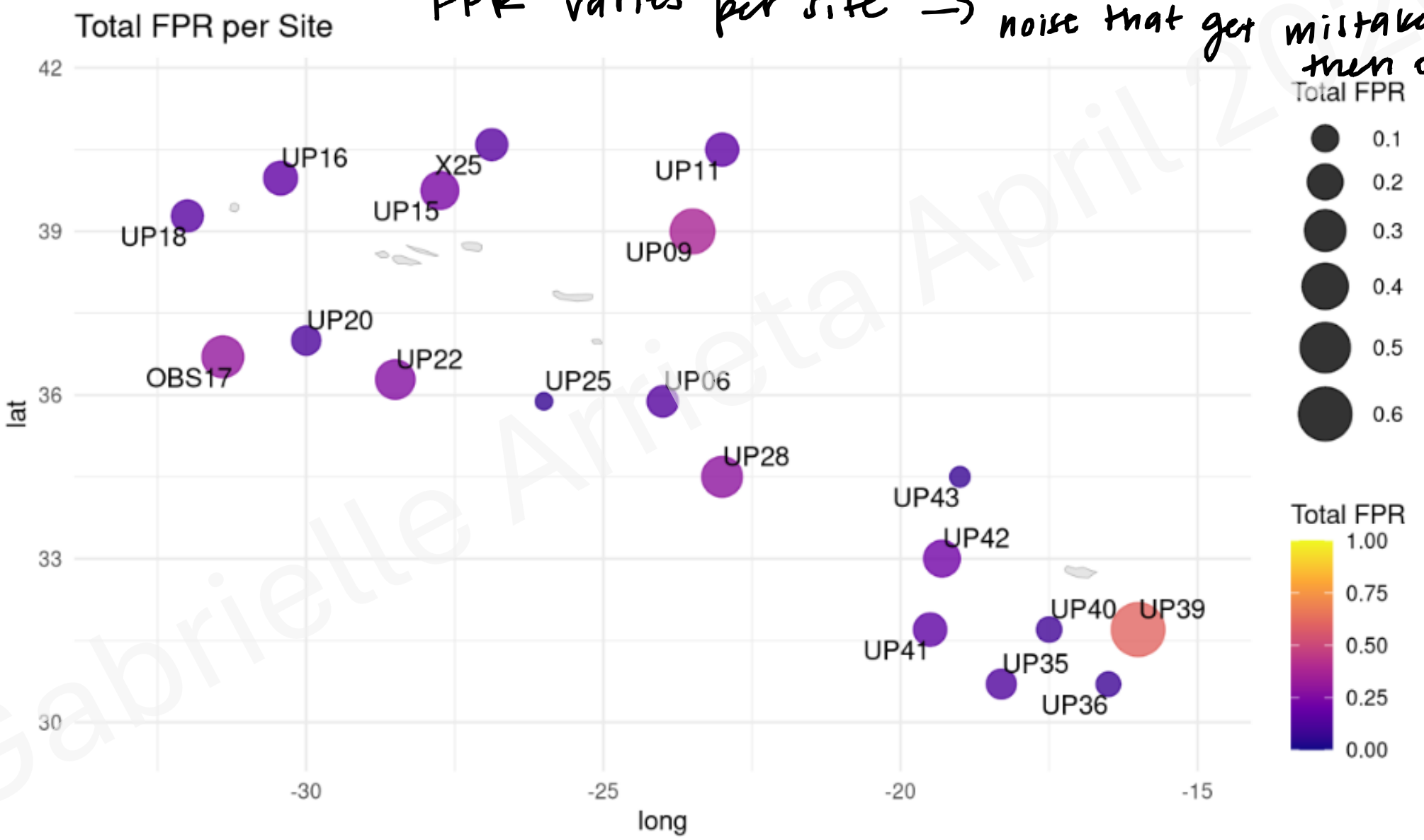
False Positive Rate Using Manual Detections

FPR varies per site → some sites have more noise that get mistaken as calls than others!

no obvious spatial trend!



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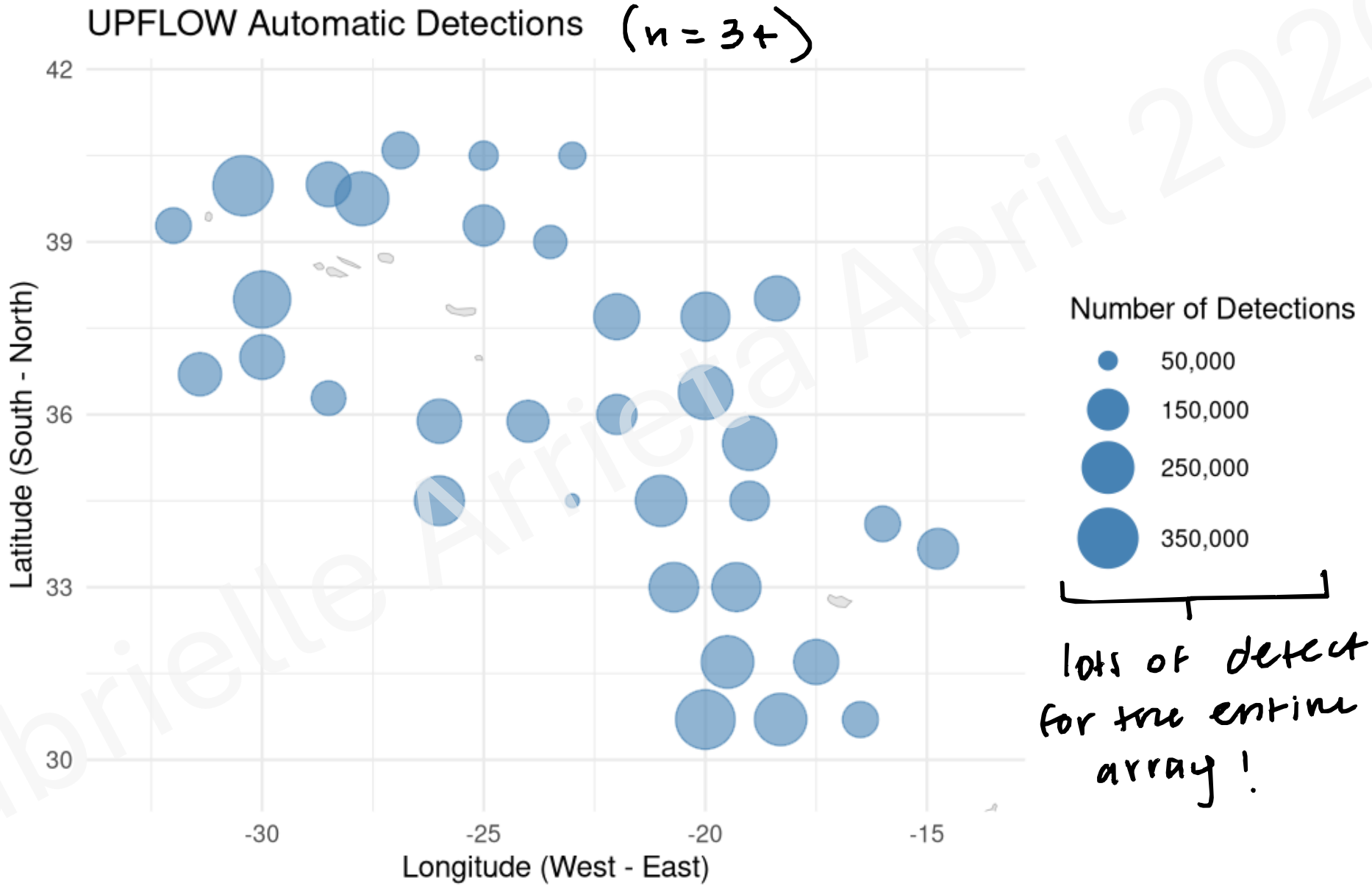
Automatic Detections → detector applied to entire dataset!



* sites w/ high FPRs were removed!

lots of detections for the entire array!

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